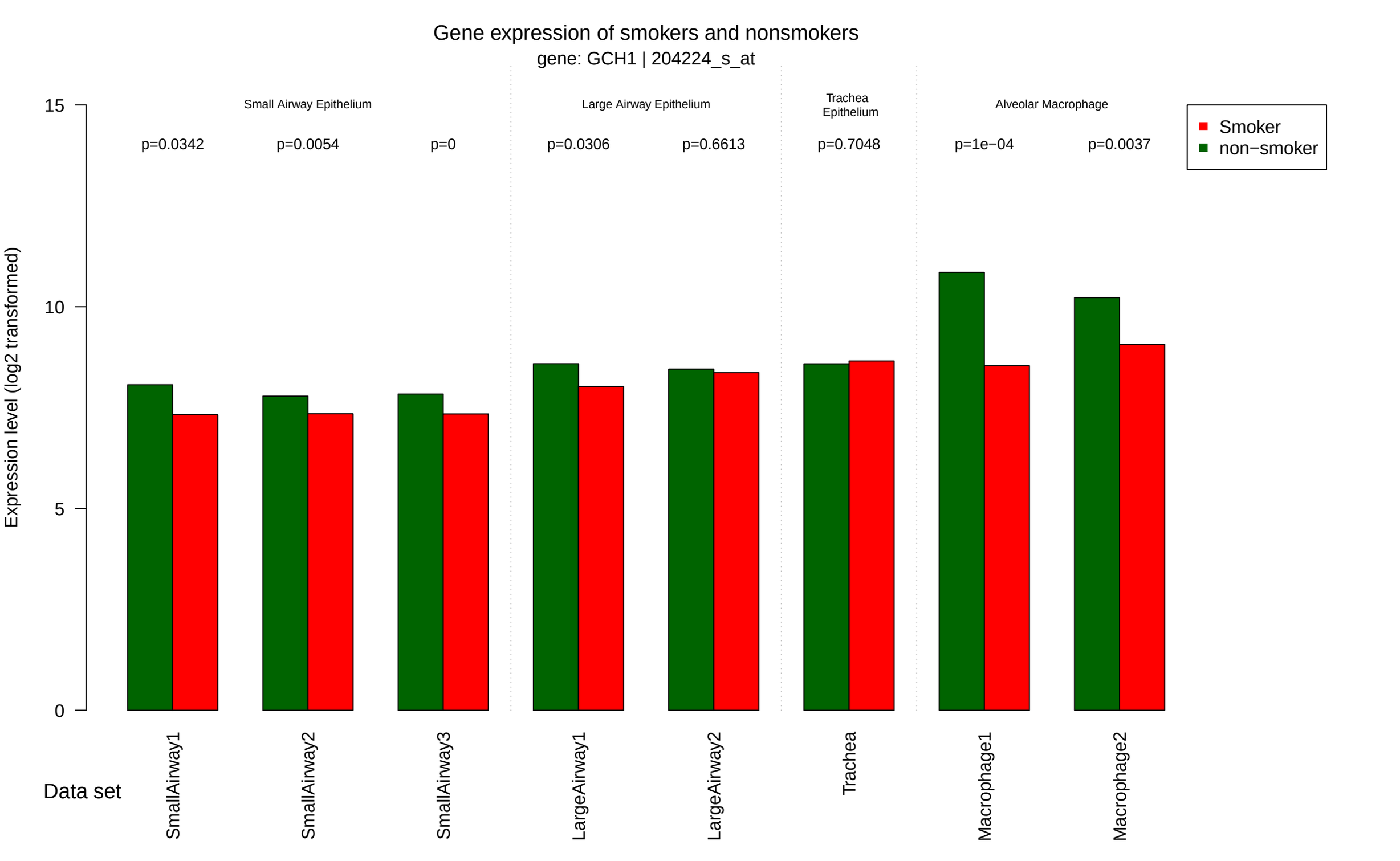
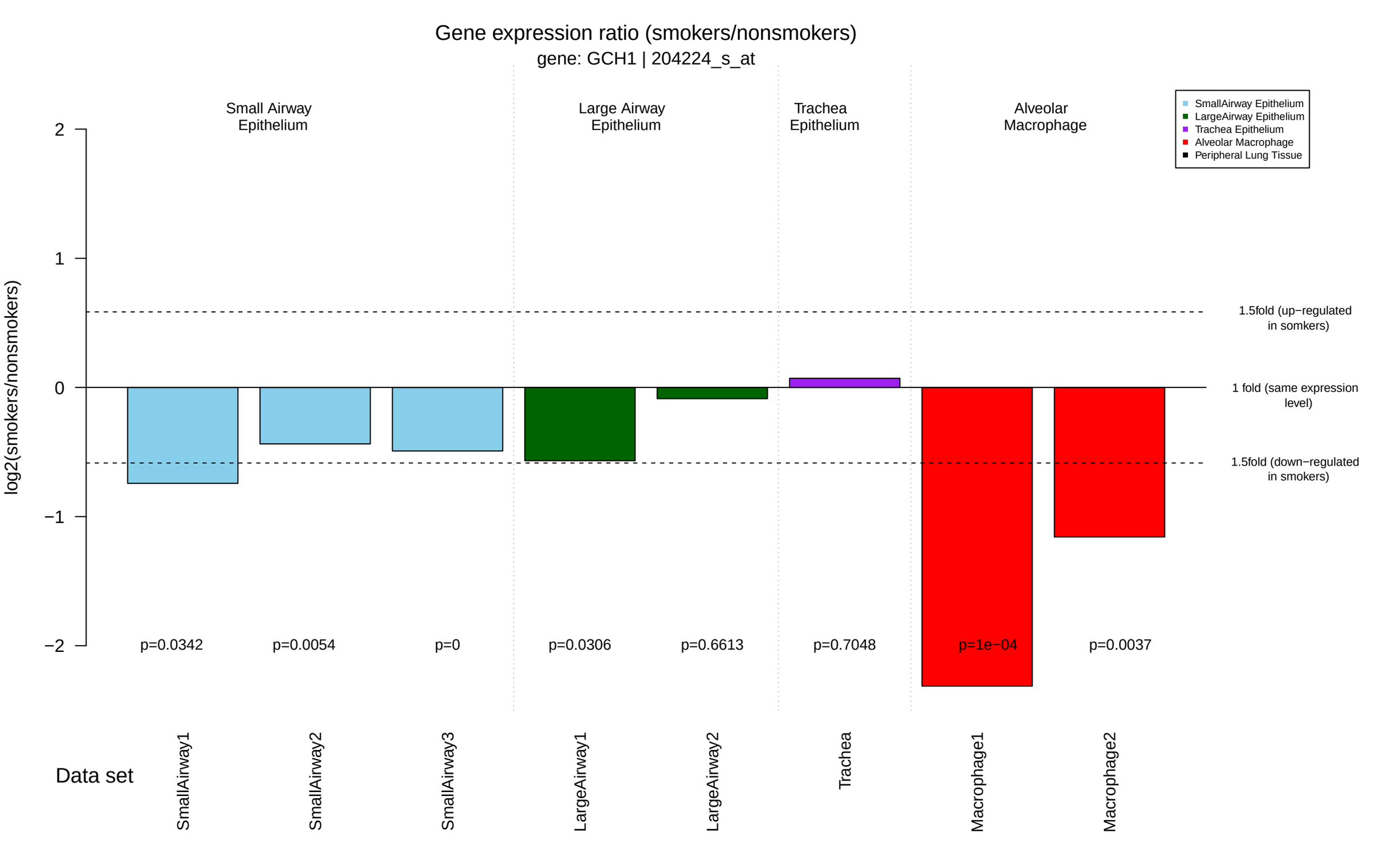
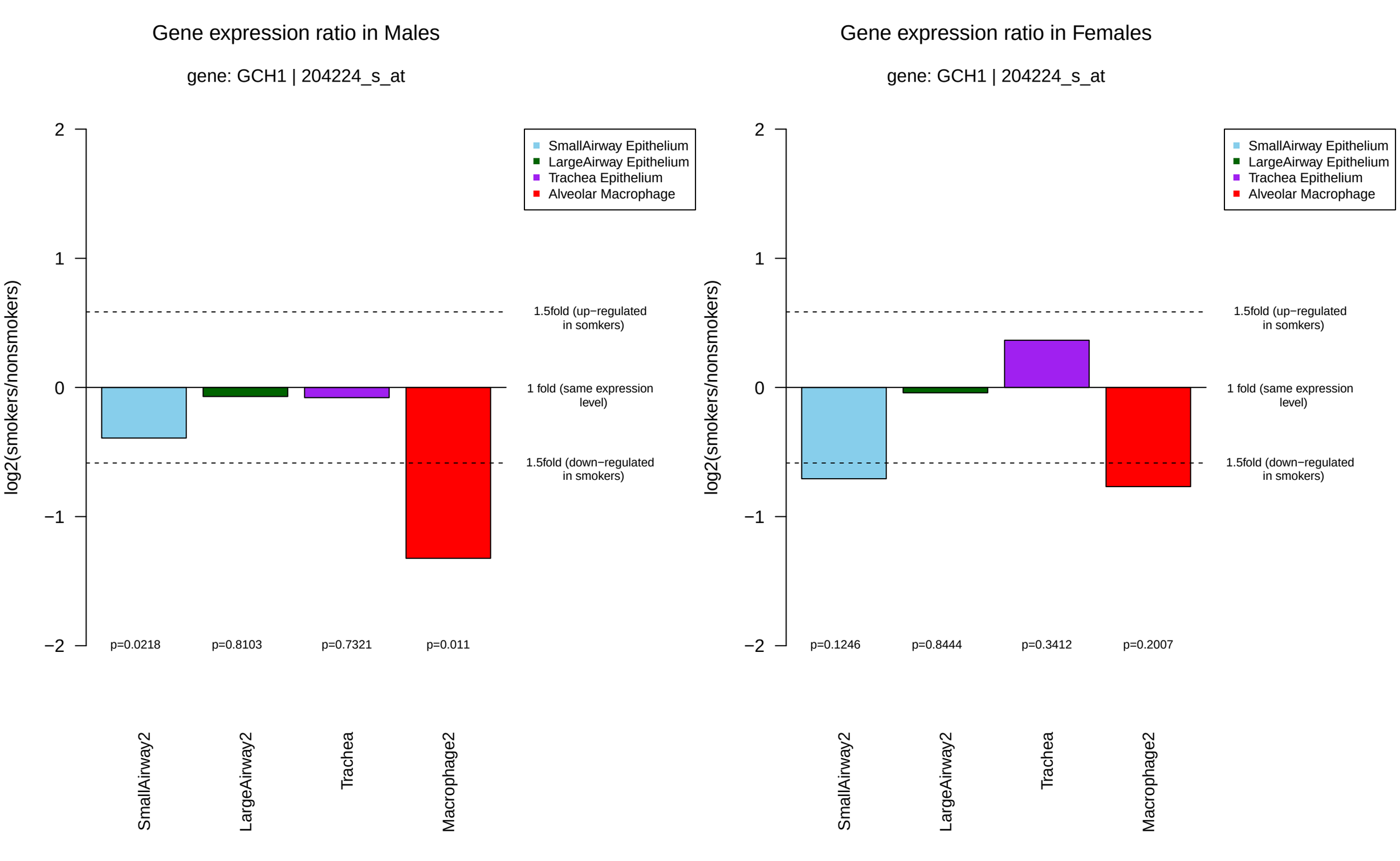
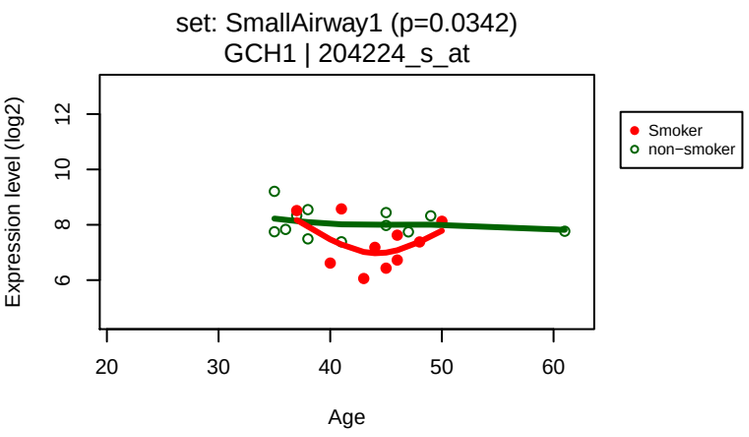
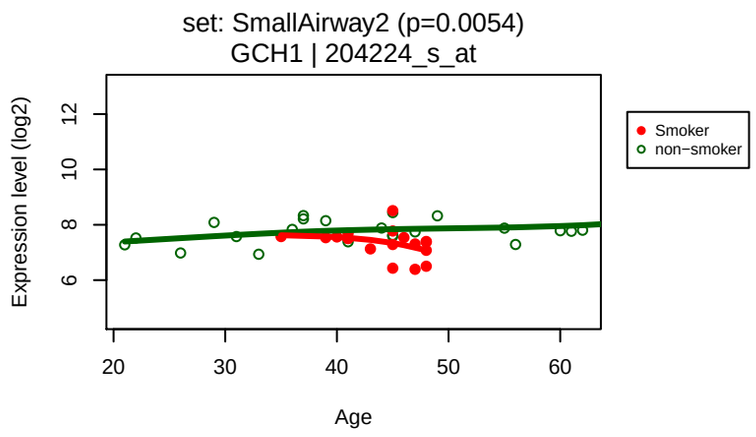
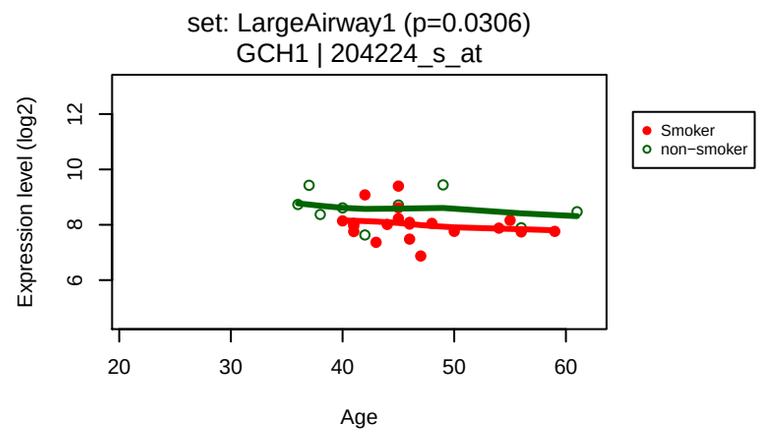

Gene expression of smokers and nonsmokers
gene: GCH1 | 204224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0342
p=0.0054
p=0
p=0.0306
p=0.6613
p=0.7048
p=1e−04
p=0.0037
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

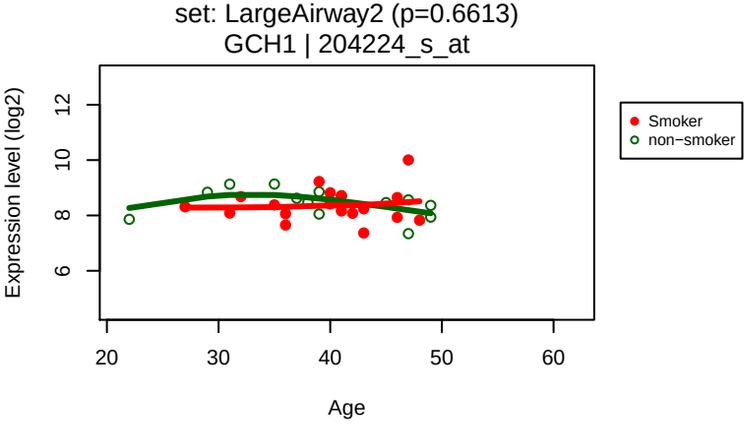

Gene expression ratio (smokers/nonsmokers)
gene: GCH1 | 204224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0342
p=0.0054
p=0
p=0.0306
p=0.6613
p=0.7048
p=1e−04
p=0.0037
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

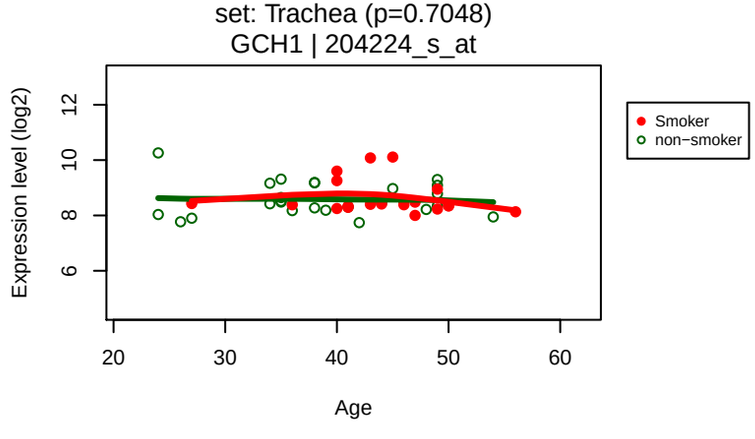

Gene expression ratio in Males
gene: GCH1 | 204224_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0218
p=0.8103
p=0.7321
p=0.011
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCH1 | 204224_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1246
p=0.8444
p=0.3412
p=0.2007
SmallAirway2
LargeAirway2
Trachea
Macrophage2

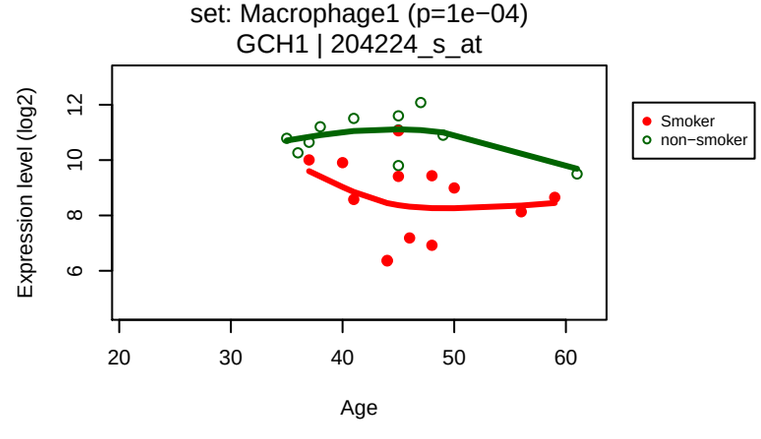

set: SmallAirway1 (p=0.0342)
GCH1 | 204224_s_at
Expression level (log2)
Age
Smoker
non−smoker

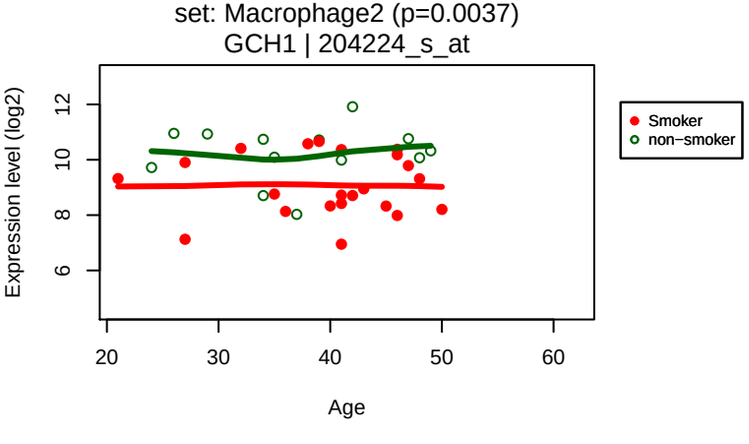

set: SmallAirway2 (p=0.0054)
GCH1 | 204224_s_at
Expression level (log2)
Age
Smoker
non−smoker

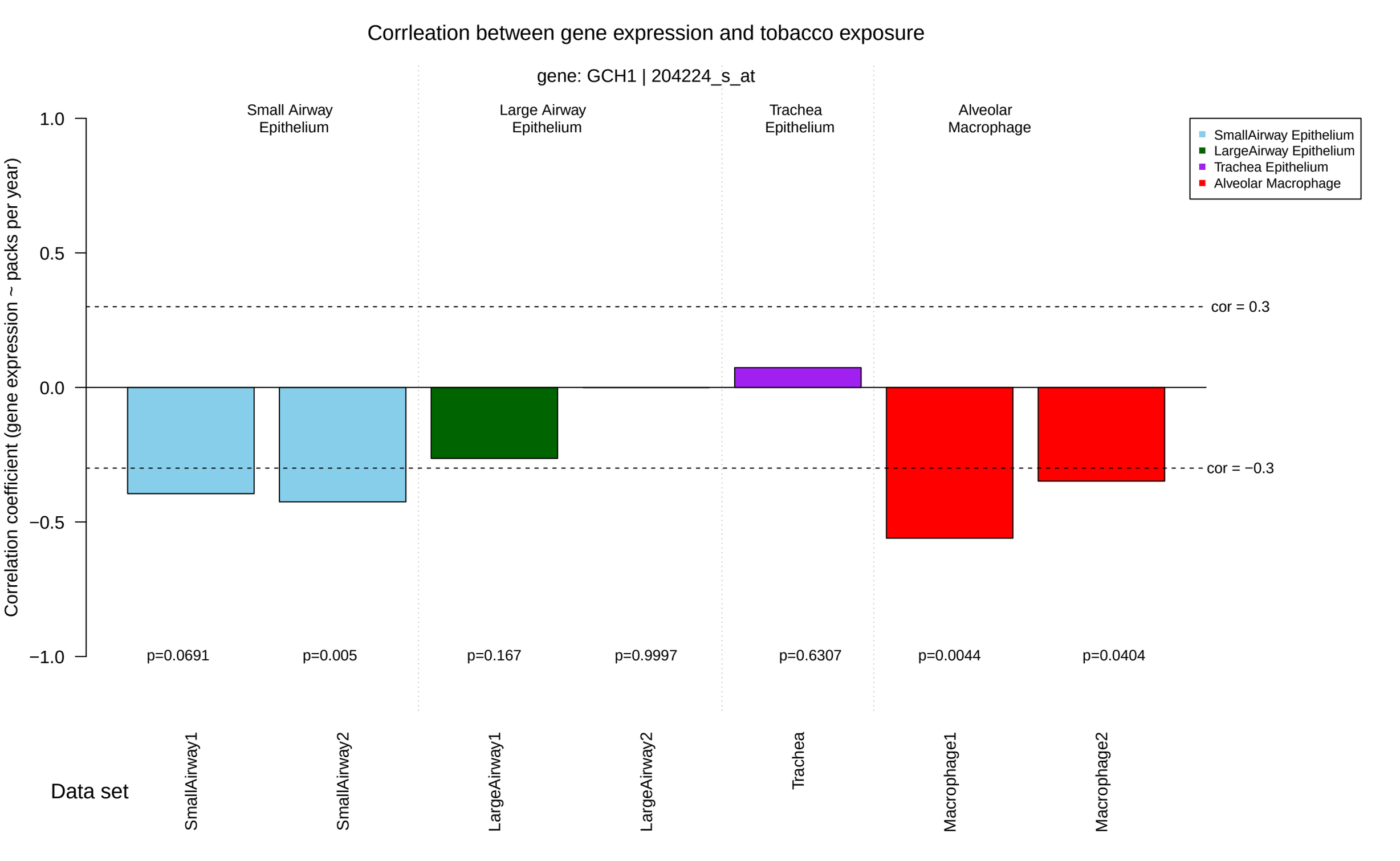

set: LargeAirway1 (p=0.0306)
GCH1 | 204224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6613)
GCH1 | 204224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7048)
GCH1 | 204224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=1e−04)
GCH1 | 204224_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0037)
GCH1 | 204224_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCH1 | 204224_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0691
p=0.005
p=0.167
p=0.9997
p=0.6307
p=0.0044
p=0.0404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2